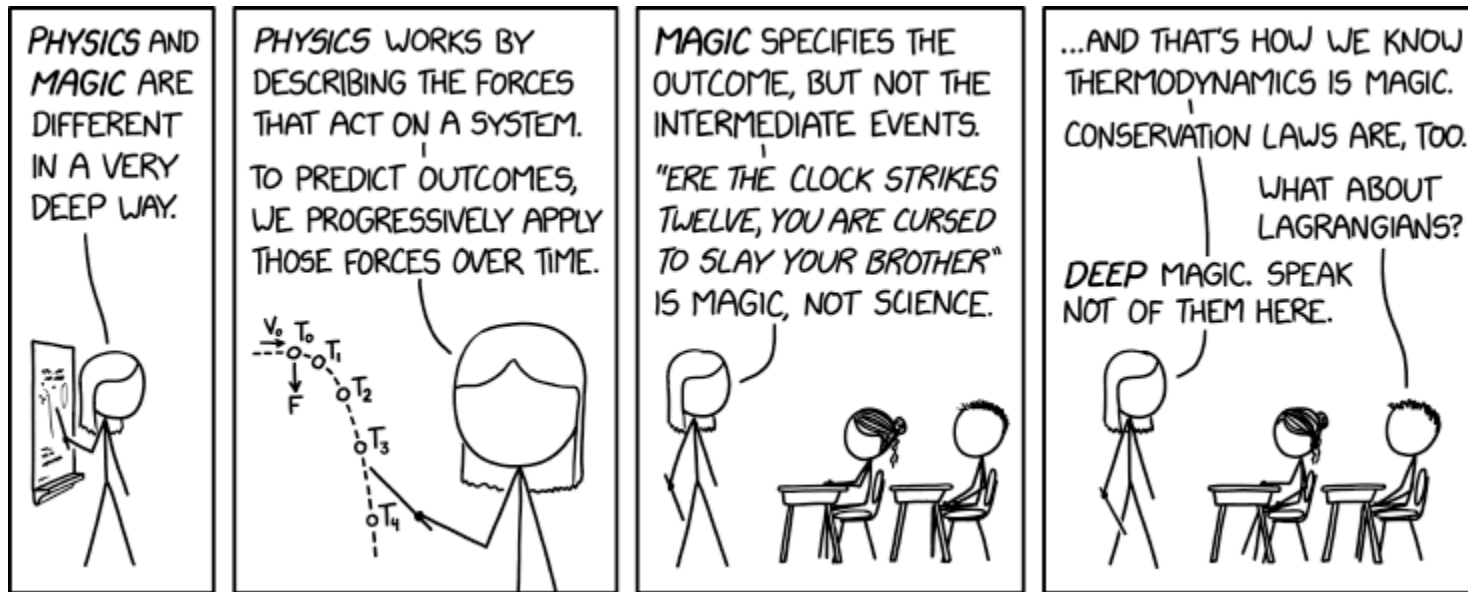




<https://xkcd.com/2904/>

Classic Mechs EP5: Non-linear Dynamics and into the Lagrangian land

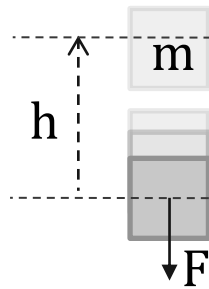
Classic Mechs EP5.1: Non-linear dynamics

Learning objectives

- 1) We will use non-linear potential field to study non-linear dynamics
- 2) We will use all the knowledge we acquired so far to linearize non-linear system at fixed point
- 3) Sometimes, we can have a very educated guess of the whole dynamic systems

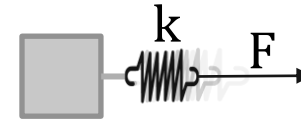
Potential energy and conservative force

The force to drop a block due to the gravity:



$$F = -\frac{\partial(mgh)}{\partial h} = -mg$$

The force from a compressed spring:



$$F = -\frac{\partial(\frac{1}{2}kx^2)}{\partial x} = -kx$$

We all agree on this intuition (in 1D): $F = -\frac{\partial U}{\partial x}$

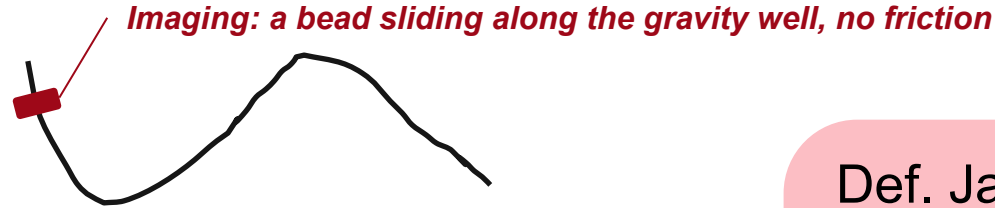
Expand it to 3D: $\vec{F}(\text{or } \underline{F}) = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}$

Or short-hand it: $\vec{F}(\text{or } \underline{F}) = -\nabla U$ (eq. 4.33)

Let's discuss this del/nabla (∇) operator later.

Non-linear potential field example:

$$U = \frac{x^2}{2} - \frac{x^3}{3}$$



1. $\underline{f} = -\frac{\partial U}{\partial \underline{x}} = ?$

2. Guess fixed points $\bar{x}$ (where $\underline{f}(\bar{x}) = 0$):

3. Linearization:

A good approxi around $\bar{x}$ with a small Δx : $\underline{f}(\underline{x}) = \frac{D\underline{f}}{D\underline{x}}(\bar{x})\Delta x$

4. Calc. “eigenstuff”, then draw phase portrait and guess!

Def. Jacobian:

$$\frac{D\underline{f}}{D\underline{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} \quad \text{When: } \underline{f} = \begin{bmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{bmatrix}$$



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Non-linear potential field example, another

$$U = \frac{x^4}{4} - \frac{x^2}{2}$$



1. $\underline{f} = -\frac{\partial U}{\partial x} = ?$

2. Guess fixed points $\bar{x}$ (where $\underline{f}(\bar{x}) = 0$):

3. Linearization:

A good approxi around $\bar{x}$ with a small Δx : $\underline{f}(x) = \frac{D\underline{f}}{Dx}(\bar{x})\Delta x$

4. Calc. “eigenstuff”, then draw phase portrait and guess!

Def. Jacobian:

$$\frac{D\underline{f}}{Dx} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} \quad \text{When: } \underline{f} = \begin{bmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{bmatrix}$$



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Classic Mechs EP5.2 The del/nabla (∇) operator

- Gradient: $\text{grad } f = \nabla f$

$$f : \text{scalar} \longrightarrow \nabla f : \text{vector}$$

- Divergence: $\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$

$$\mathbf{F} : \text{vector} \longrightarrow \nabla \cdot \mathbf{F} : \text{scalar}$$

- Curl: $\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$

$$\mathbf{F} : \text{vector} \longrightarrow \nabla \times \mathbf{F} : \text{vector}$$

The del/nabla (∇) operator

<https://en.wikipedia.org/wiki/Del>

- Gradient: $\text{grad } f = \nabla f$

$$\nabla f = \left(\mathbf{e}_x \frac{\partial}{\partial x} \right) f + \left(\mathbf{e}_y \frac{\partial}{\partial y} \right) f + \left(\mathbf{e}_z \frac{\partial}{\partial z} \right) f = \frac{\partial f}{\partial x} \mathbf{e}_x + \frac{\partial f}{\partial y} \mathbf{e}_y + \frac{\partial f}{\partial z} \mathbf{e}_z$$

- Divergence: $\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$

$$\nabla \cdot \mathbf{F} = \left(\mathbf{e}_x \frac{\partial}{\partial x} \cdot \mathbf{F} \right) + \left(\mathbf{e}_y \frac{\partial}{\partial y} \cdot \mathbf{F} \right) + \left(\mathbf{e}_z \frac{\partial}{\partial z} \cdot \mathbf{F} \right) = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

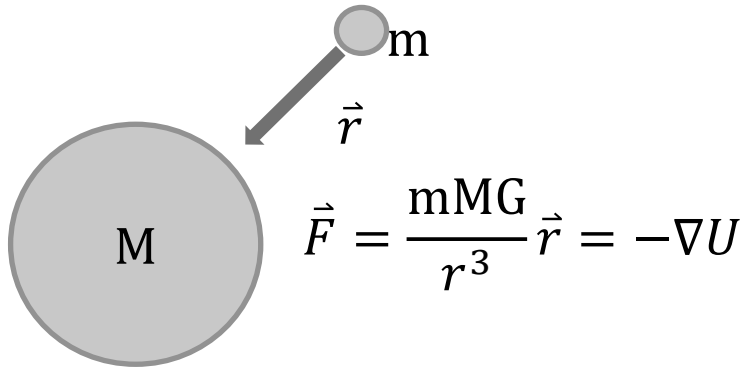
- Curl: $\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

$$\begin{aligned} \nabla \times \mathbf{F} &= \left(\mathbf{e}_x \frac{\partial}{\partial x} \times \mathbf{F} \right) + \left(\mathbf{e}_y \frac{\partial}{\partial y} \times \mathbf{F} \right) + \left(\mathbf{e}_z \frac{\partial}{\partial z} \times \mathbf{F} \right) \\ &= \frac{\partial}{\partial x} (0, -F_z, F_y) + \frac{\partial}{\partial y} (F_z, 0, -F_x) + \frac{\partial}{\partial z} (-F_y, F_x, 0) \end{aligned}$$

$$= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \mathbf{e}_x + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \mathbf{e}_y + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \mathbf{e}_z$$

Example: object in a gravitational field:



In cartesian:

$$U = \frac{mMG}{\sqrt{x^2 + y^2 + z^2}}$$

$$\frac{\partial U}{\partial x} = \frac{mMG}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} x = \frac{mMG}{r^3} x$$

$$\frac{\partial U}{\partial y} = \frac{mMG}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} y = \frac{mMG}{r^3} y$$

$$\frac{\partial U}{\partial z} = \frac{mMG}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} z = \frac{mMG}{r^3} z$$

$$\vec{F} = -\nabla U = \begin{bmatrix} -\frac{mMGx}{r^3} \\ -\frac{mMGy}{r^3} \\ -\frac{mMGz}{r^3} \end{bmatrix}$$

And also: how is the gravitational force linearized on earth surface assuming $R(\text{earth radius}) \gg h(\text{height})$?



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